Database Development and Design (SOE201)

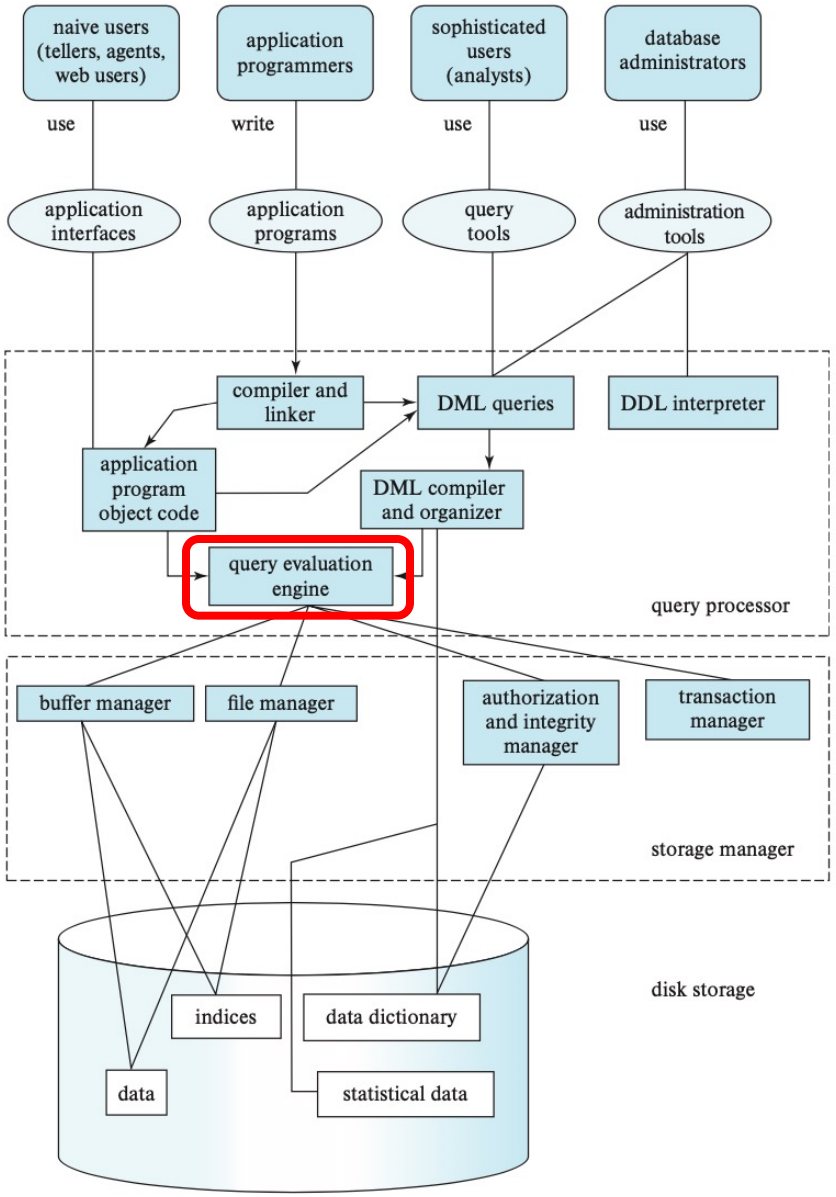
Lecture 4b - Query Evaluation Basics

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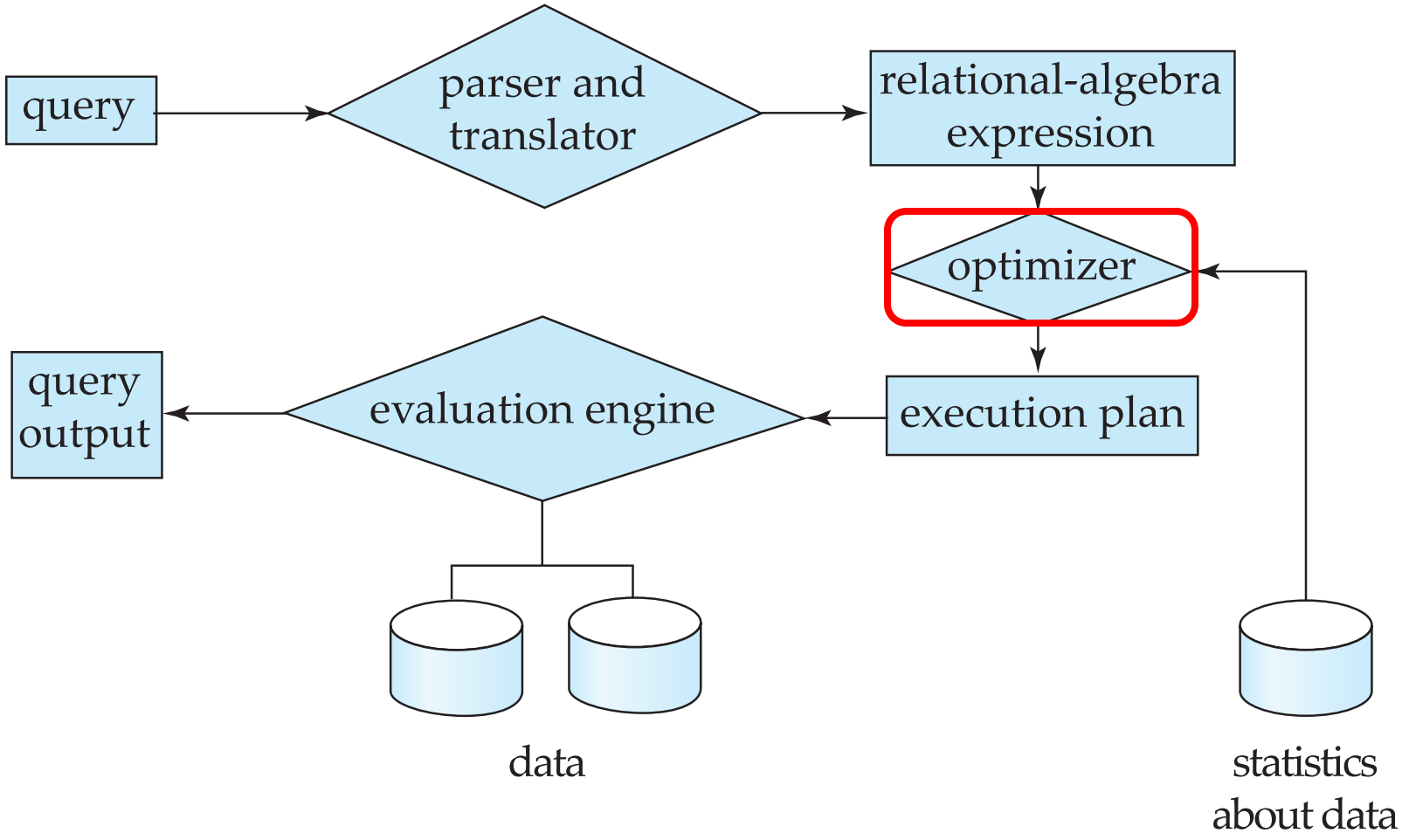
Learning Outcomes

- Query processing basics
  - Basic steps in query processing
  - Measure query costs

DBMS Components



Query Processing



Basic Steps in Query Processing

- Three basic steps in query processing:
  - Parsing and translation
  - Optimisation
  - Evaluation

Parsing and Translation

- Translate query into its internal form
- Then translate into relational algebra expression
- Parser checks syntax and verifies relations
  - This is subject for *compilers*

Evaluation

- Query execution engine
  - Takes evaluation plan
  - Executes that plan
  - Returns answers to query
- However, bulk of problem lies in *how* to come up with good evaluation plans.
  - *Query optimisation !*

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Optimisation

- Relational algebra expression may have multiple equivalent expressions, e.g.
  - $\sigma_{balance<2500}(\Pi_{balance}(account))$ VS $\Pi_{balance}(\sigma_{balance<2500}(account))$
- Each relational algebra operation can be evaluated using different algorithms
  - Correspondingly, relational algebra expression can be evaluated in many ways.
- Annotated expression specifying detailed evaluation strategy is called evaluation plan, e.g.
  - Plan 1: use index on *balance* to find account balance < 2500
  - Plan 2: use linear scan and discard account balance ≥ 2500

Optimisation – cont'd

- **Query Optimisation**: amongst all equivalent evaluation plans, choose one with lowest cost.
  - Cost is estimated using **statistical information** from database catalog
    - e.g. number of tuples in each relation, size of tuples, number of blocks in relation, etc.
- Issues to cover:
  - How to measure query costs
  - What are algorithms for evaluating individual relational algebra operations
  - How to combine algorithms for individual operations to evaluate complex expressions
  - How to optimise queries, that is, how to find evaluation plan with lowest estimated cost

Measures of Query Cost

- Cost of query evaluation can be measured in terms of different resources
  - *Disk access*
  - CPU cost
    - Ignore CPU costs for simplicity; real systems do take CPU cost into account
  - Network communication
    - Network costs must be considered for parallel systems

Measures of Query Cost – cont'd

- **Disk cost** can be estimated as:
  - Number of seeks (average-seek-cost)
  - Number of blocks read (average-block-read-cost)
  - Number of blocks written (average-block-write-cost)
- For simplicity, use **number of block transfers** *from/to disk and* **number of seeks** as cost measures
  - *Assume for simplicity that write cost is 'same' as read cost*
  - t_T – time to transfer one block
  - t_S – time for one seek
  - Cost for **b** block transfers plus **s** seeks

    $b * t_T + s * t_S$
- t_S and t_T depend on where/how data is stored; with 4 KB blocks:
  - High-end magnetic disk: t_S = 4 msec and t_T =0.1 msec
  - SSD: t_S = 20-90 microsec and t_T = 2-10 microsec

msec (millisecond or ms); microsec (microsecond or μs); 1ms=1,000μs

End of Lecture

- Reading
  - 7th edition, Chapters 15.1 and 15.2